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Autoregressive Moving Average Process (ARMA)

Course: Practical Time Series Analysis

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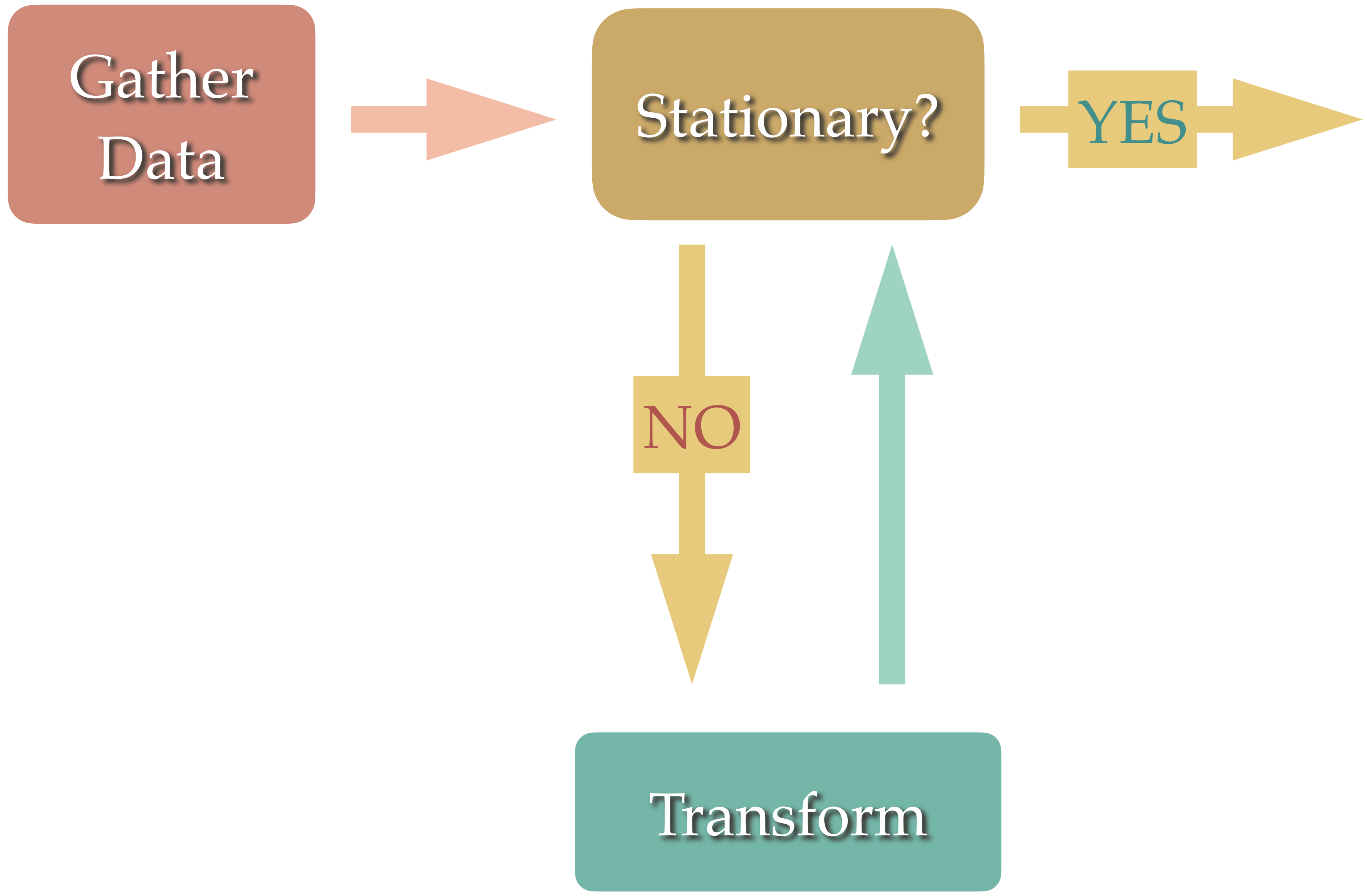

Identifying an Autoregressive Moving Average Process

If the process is stationary and *both the ACF and PACF plots show a decaying or sinusoidal pattern*, then it is a stationary **ARMA(p, q) process**.

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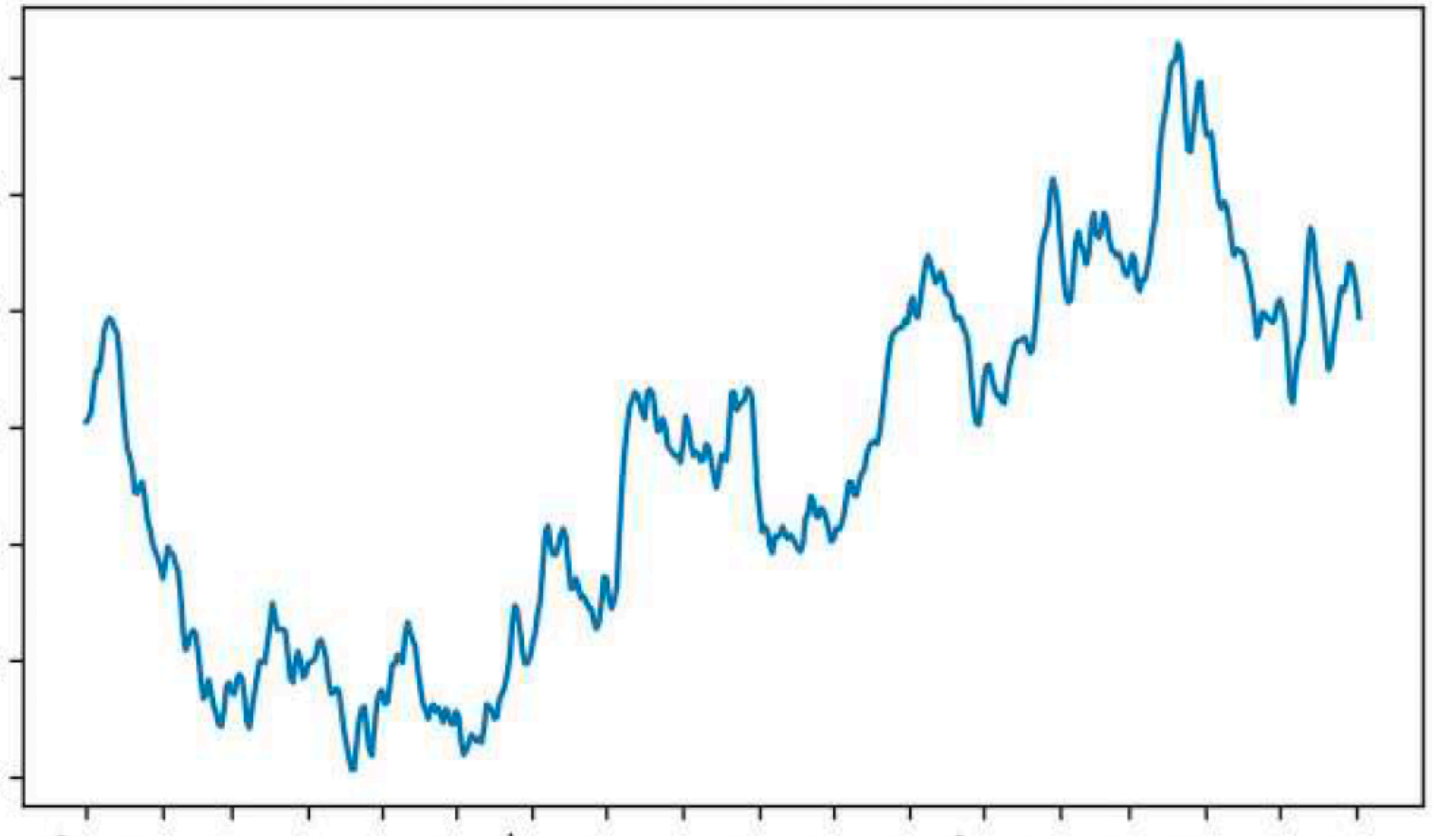
01

Autoregressive Moving Average Process (ARMA)

$\stackrel{\text{def}}{=}$ What is a Moving Average Process?

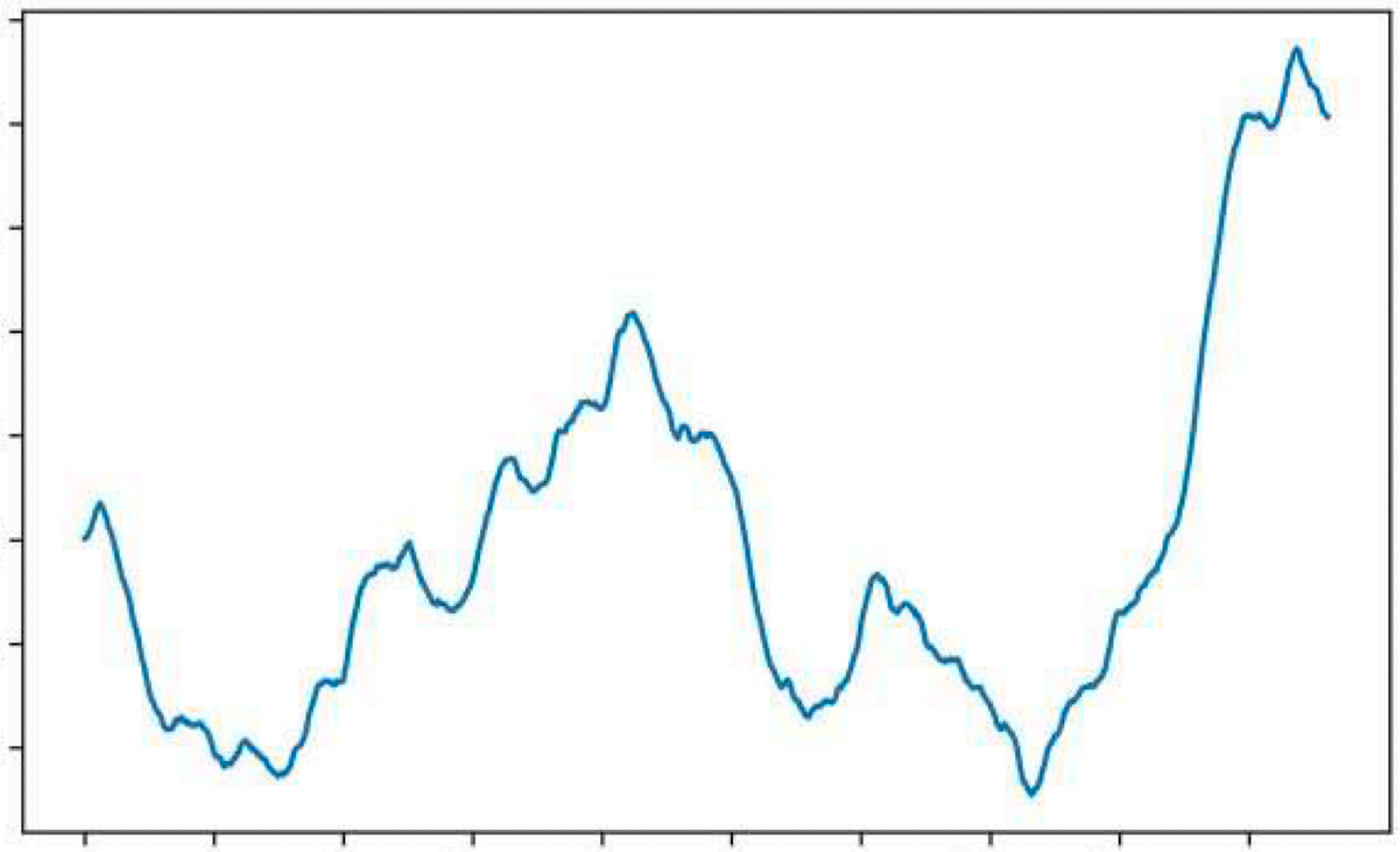


- In a **moving average (MA) process**, the current value depends linearly on the mean of the series, the current error term, and past error terms.
- The moving average model is denoted as MA(q), where q is the **order**. The general expression of an MA(q) model is

$$y_t = \mu + \epsilon_t + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \cdots + \theta_q \epsilon_{t-q} \,.$$

- The error terms are assumed to be mutually *independent* and *normally distributed*, just like white noise.
- The magnitude of the impact of past errors on the present value is quantified using a coefficient.

≝ What is an Autoregressive Process?



- An **autoregressive process** is a regression of a variable against itself. In a time series, this means that the present value is linearly dependent on its past values.
- The autoregressive process is denoted as AR(p), where p is the **order**. The general expression of an AR(p) model is

$$y_t = C + \epsilon_t + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p},$$

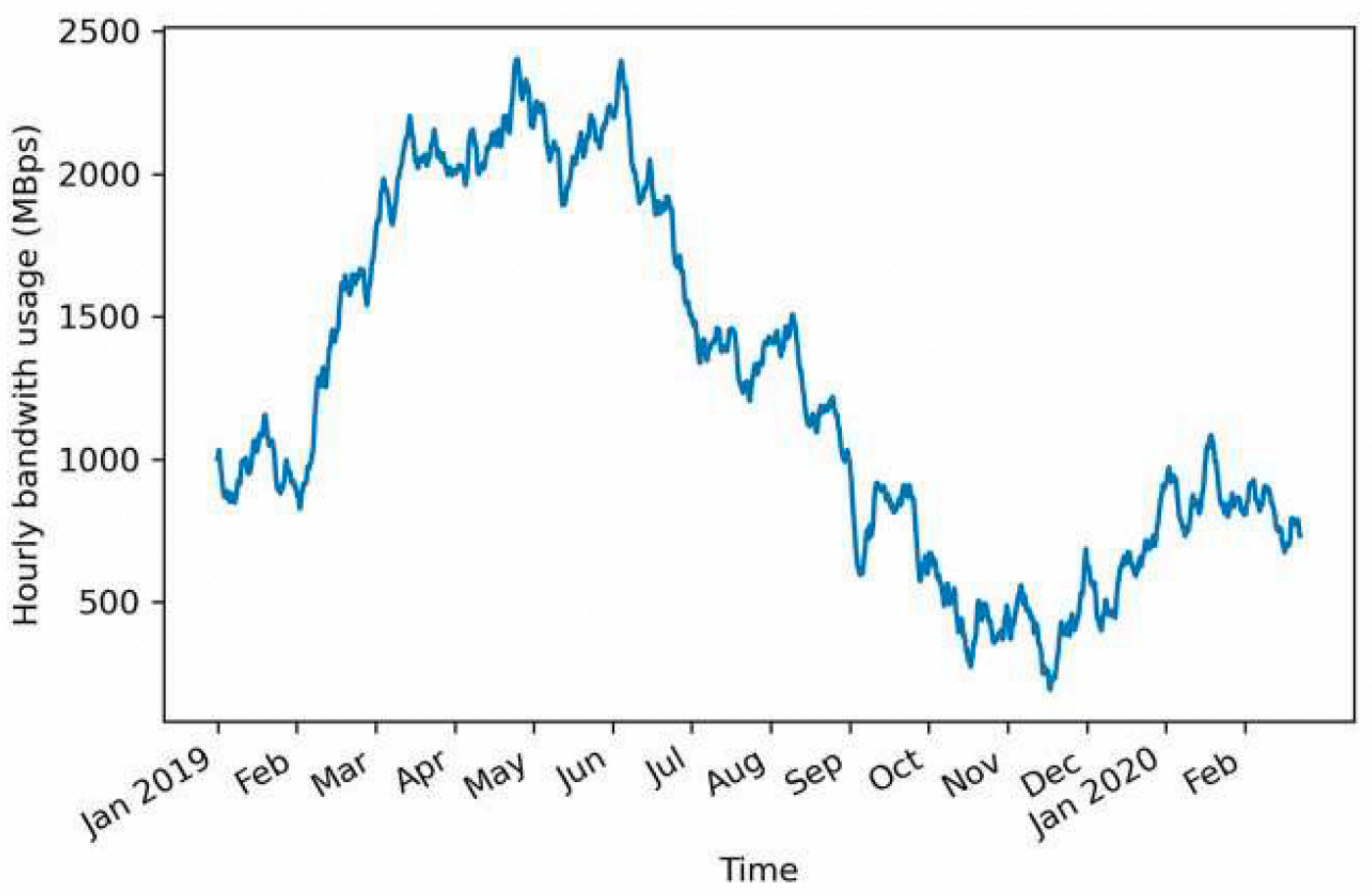
where C is a constant, ϵ_t is an error (white noise) and ϕ_t are coefficients (magnitude of influence of past values).

$\stackrel{\text{def}}{=}$ What is an Autoregressive Moving Average Process (ARMA)?

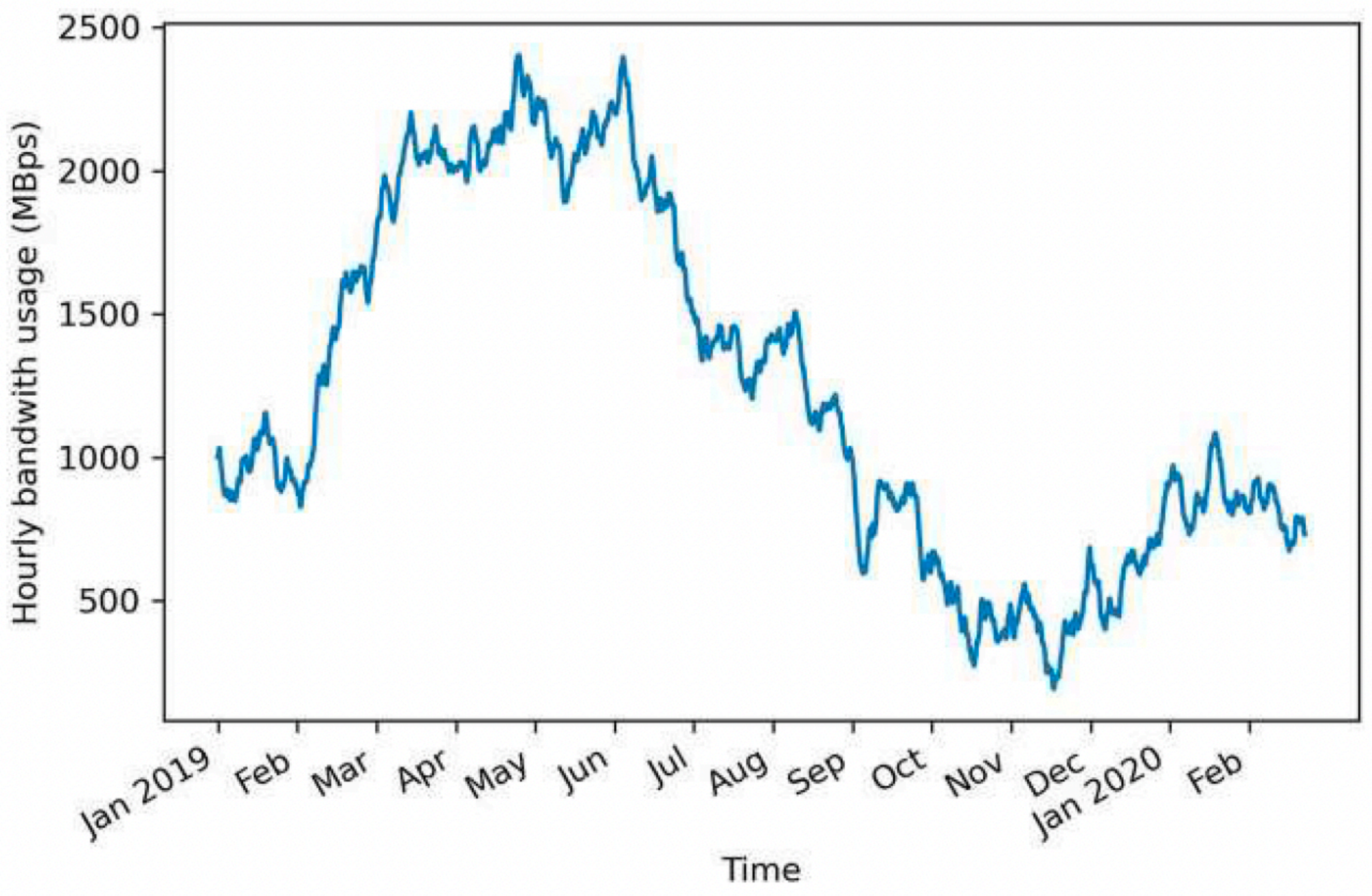
- The **autoregressive moving average process** is a combination of the autoregressive process and the moving average process.
- It is denoted by ARMA(p, q), where is the **order** p of the autoregressive part, and q is the **order** of the moving average part. The general expression is

$$y_t = C + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + \mu + \epsilon_t + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \cdots + \theta_q \epsilon_{t-q}$$

Example 1: Hourly bandwidth usage in a data centre since January 1, 2019 (10,000 points)



Example 1: Hourly bandwidth usage in a data centre since January 1, 2019 (10,000 points)



- Long-term trends.
- Not stationary.
- No apparent cyclical pattern.

Example 2

1. What is ARMA$(0, q)$?
2. What is ARMA$(p, 0)$?
3. What is ARMA$(1, 0)$?

- The general expression of ARMA(p, q) is

$$y_t = C + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p}$$
$$+\mu + \epsilon_t + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \cdots + \theta_q \epsilon_{t-q}$$

Example 2

1. What is ARMA$(0, q)$?
2. What is ARMA$(p, 0)$?
3. What is ARMA$(1, 0)$?

Answers:

1. MA(q)
2. AR(p)
3. random walk

- The general expression of ARMA(p, q) is

$$y_t = C + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} \\ + \mu + \epsilon_t + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \cdots + \theta_q \epsilon_{t-q}$$

Example 3: ARMA(1, 1) — Simulation

$$y_t = 0.33y_{t-1} + \epsilon_t + 0.9\epsilon_{t-1}$$

- Import the **ArmaProcess** function from the **statsmodels** library:

```
from statsmodels.tsa.arima_process import ArmaProcess
import numpy as np

np.random.seed(42)

ar1 = np.array([1, -0.33]

ma1 = np.array([1, 0.9])

ARMA_1_1 = ArmaProcess(ar1, ma1).generate_sample(nsample=1000)
```

Example 3: ARMA(1, 1) — Stationarity

$$y_t = 0.33y_{t-1} + \epsilon_t + 0.9\epsilon_{t-1}$$

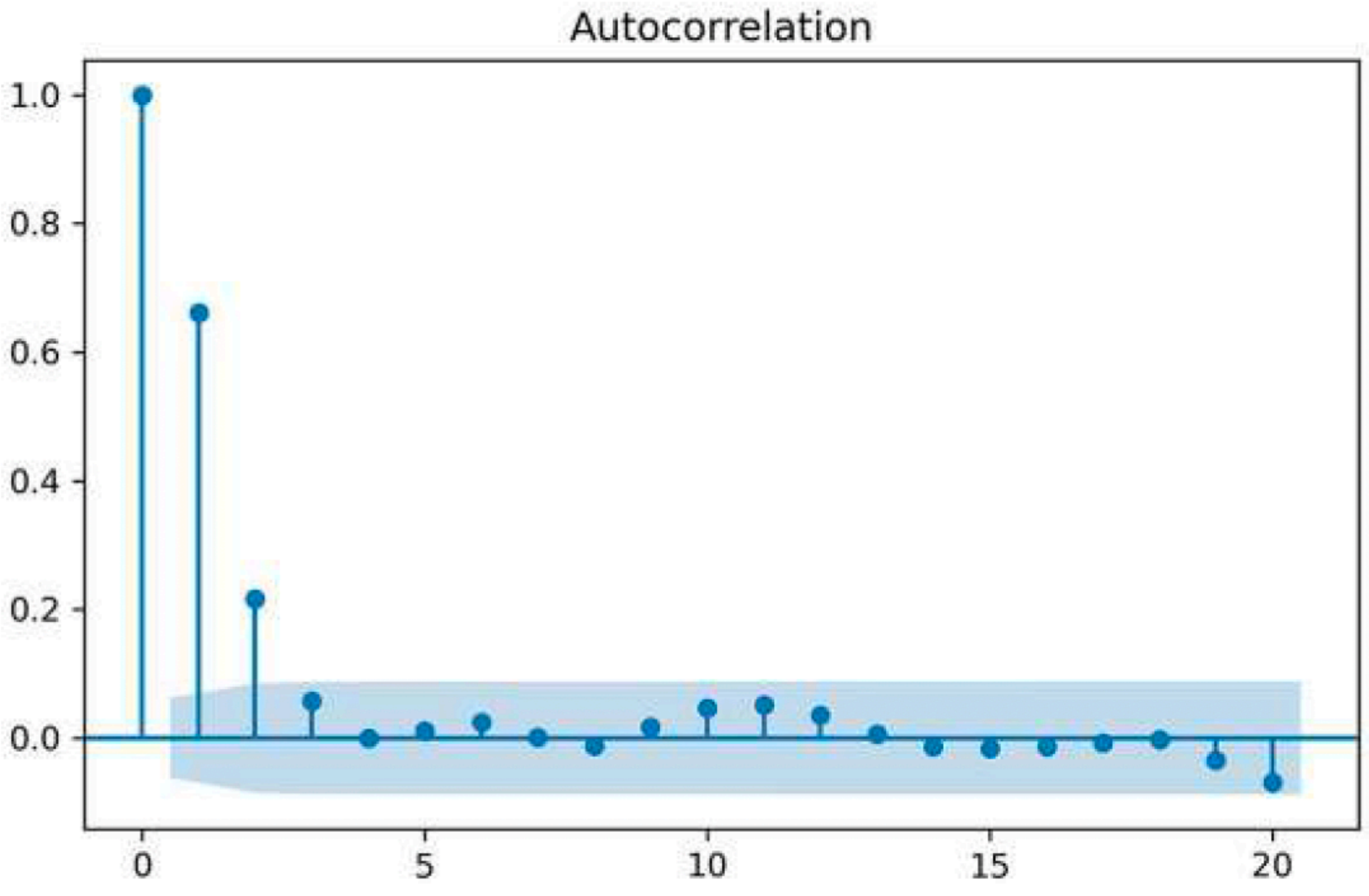
- Use the **adfuller** function from **statsmodels**:

```
from statsmodels.tsa.stattools import adfuller
ADF_result = adfuller(ARMA_1_1)
print(f'ADF Statistic: {ADF_result[0]}')
print(f'p-value: {ADF_result[1]}')
```

This results in an ADF statistic of –6.43 and a p-value of 1.7×10^{-8}.
Stationary.

Example 3: ARMA(1, 1) — Autocorrelation

$$y_t = 0.33y_{t-1} + \epsilon_t + 0.9\epsilon_{t-1}$$



- The next step is to check whether the series is *autocorrelated*. Use the **plot_acf** function from the statsmodels library:

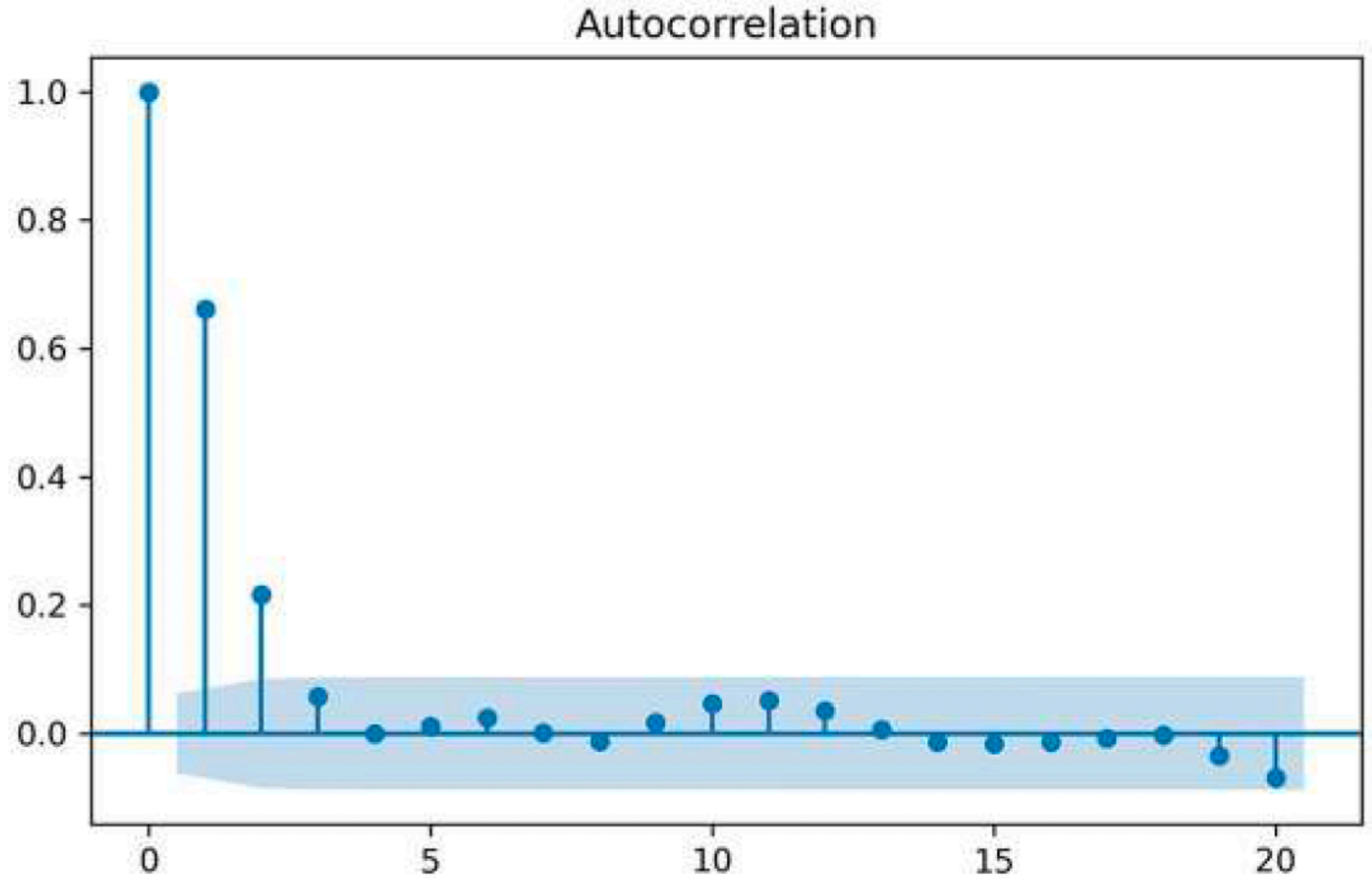
  from statsmodels.graphics.tsaplots import plot_acf

  plot_acf(ARMA_1_1, lags=20)

- There is a sinusoidal pattern on the plot, meaning that an AR(p) process is in play.
- Also, the last significant coefficient is at lag 2, which suggests that $q = 2$. However, we know that we simulated an ARMA(1, 1) process, so q must be equal to 1!
- So, the ACF plot does not reveal any useful information about the order q.

Example 3: ARMA(1, 1) — Partial Autocorrelation

$$y_t = 0.33y_{t-1} + \epsilon_t + 0.9\epsilon_{t-1}$$



- Use the **plot_pacf** function from **statsmodels**:

  from statsmodels.graphics.tsaplots import plot_pacf

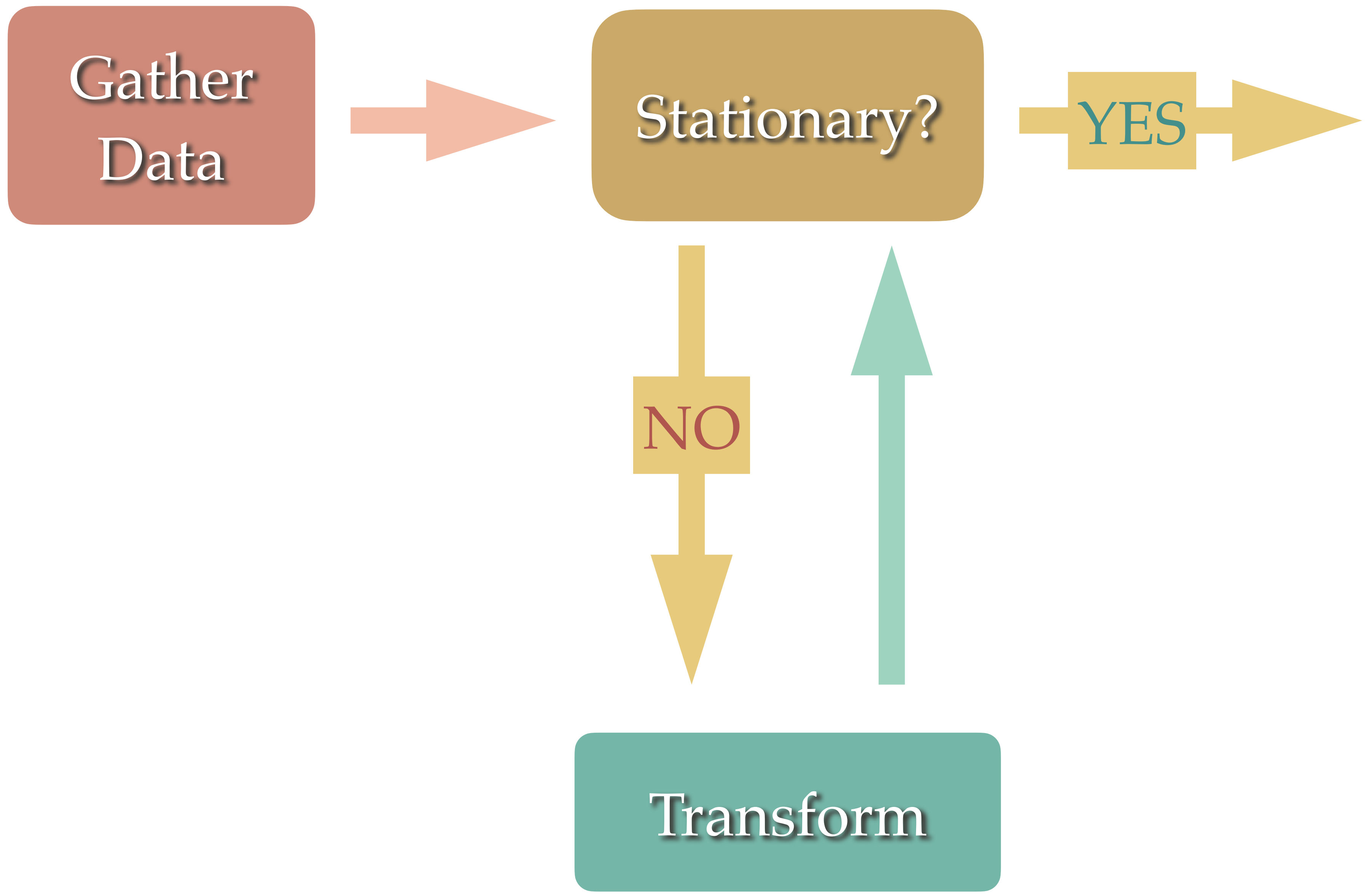
  plot_pacf(ARMA_1_1, lags=20)

- There is a sinusoidal pattern with no clear cutoff between significant and non-significant coefficients. We cannot infer that $p = 1$.
- So, the PACF plot does not reveal any useful information about the order p.

02

General Modelling Procedure

Identifying an Autoregressive Moving Average Process



Identifying an Autoregressive Moving Average Process

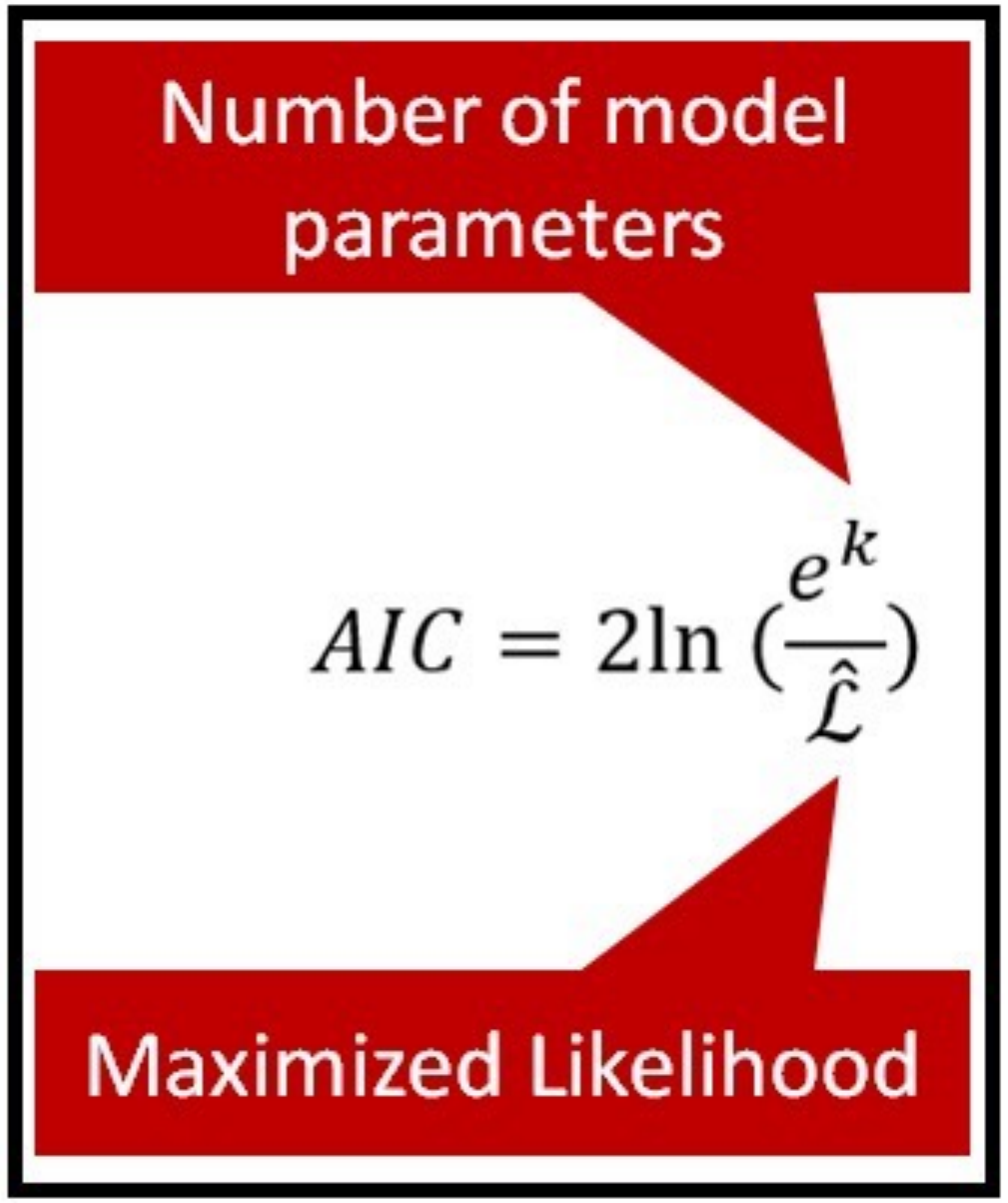
Why the general procedure is better?

- It allows us to select a model based entirely on statistical tests and numerical criteria, instead of relying on the qualitative analysis of the ACF and PACF plots.
- It can also be applied in situations where our time series is non-stationary and has seasonal effects.

03

Akaike Information Criterion (AIC)

≝ What is Akaike information criterion (AIC)?

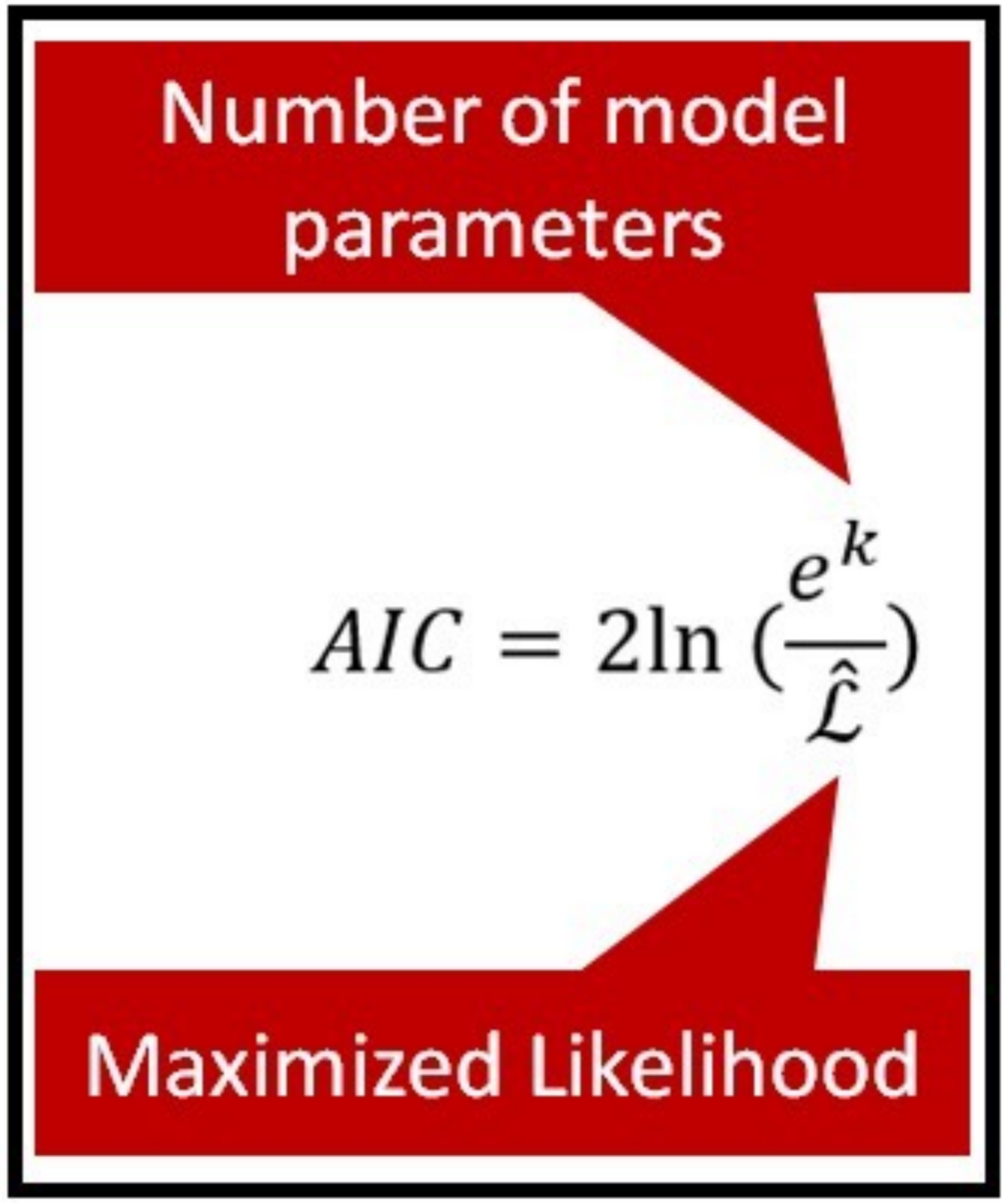


- The **Akaike information criterion (AIC)** is a measure of the quality of a model in relation to other models. It is used for model selection.
- The AIC is a function of the number of parameters k in a model and the maximum value of the likelihood function $\hat{L}$:

$$AIC = 2k - 2\ln(\hat{L})$$

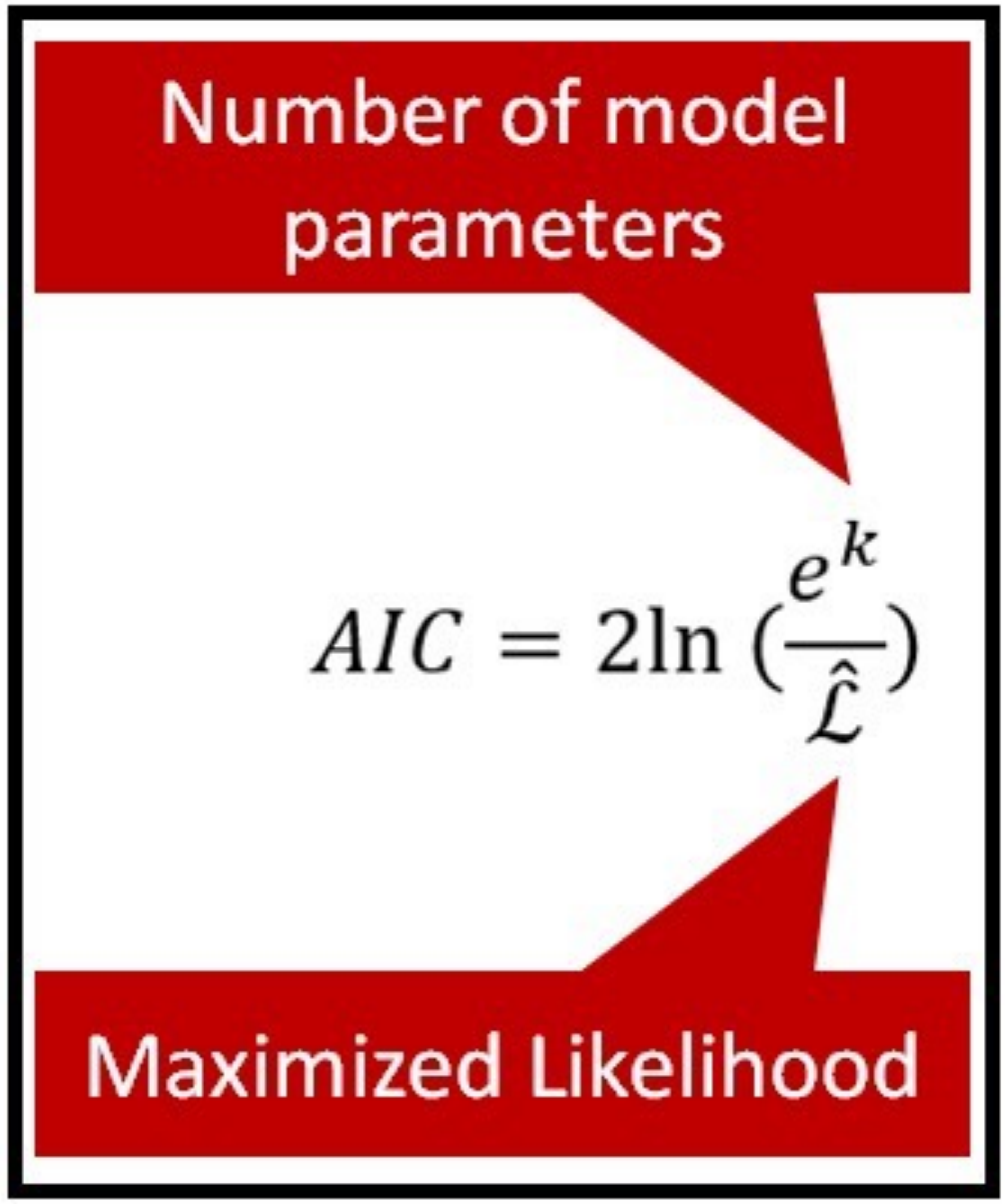
- The lower the value of the AIC, the better the model. Selecting according to the AIC allows us to keep a balance between the complexity of a model and its goodness of fit to the data.

What is Akaike information criterion (AIC)?



- The **number of estimated parameters** k is directly related to the order (p, q) of an ARMA(p, q) model. If we fit an ARMA(2, 2) model, then we have $2 + 2 = 4$ parameters to estimate. If we fit an ARMA(3, 4) model, then we have $3 + 4 = 7$ parameters to estimate.
- The **likelihood function** measures the goodness of fit of a model. It can be viewed as the opposite of the distribution function. Given a model with fixed parameters, the distribution function will measure the probability of observing a data point. The likelihood function flips the logic. Given a set of observed data, it will estimate how likely it is that different model parameters will generate the observed data.

≝ What is Akaike information criterion (AIC)?



- The AIC quantifies the quality of a model in relation to other models only. It is therefore a *relative measure of quality*.
- In the event that we fit only poor models to our data, the AIC will simply help us determine the best from that group of models.

A bit of history…

- **Hirotsugu Akaike** (November 5, 1927 – August 4, 2009) was a Japanese statistician.
- He formulated the Akaike information criterion (AIC). AIC is now widely used for model selection, which is commonly the most difficult aspect of statistical inference.
- It was first announced in English by Akaike at a 1971 symposium. The first formal publication was a 1974 paper by Akaike. As of October 2014, the 1974 paper had received more than 14,000 citations in the Web of Science: making it the 73rd most-cited research paper of all time.
- Akaike also made major contributions to the study of time series. As well, he had a large role in the general development of statistics in Japan.

Example 3: ARMA(1, 1) — AIC

$$y_t = 0.33y_{t-1} + \epsilon_t + 0.9\epsilon_{t-1}$$

- Allow the values of p and q to vary from 0 to 3 (this range is arbitrary). Create a list of all possible combinations of (p, q), using the **product** function from **itertools**:

```
from itertools import product

ps = range(0, 4, 1)
qs = range(0, 4, 1)
order_list = list(product(ps, qs))
```

Example 3: ARMA(1, 1) — AIC

$$y_t = 0.33y_{t-1} + \epsilon_t + 0.9\epsilon_{t-1}$$

- Fit all unique ARMA models:

```
from typing import Union
from tqdm import tqdm_notebook
from statsmodels.tsa.statespace.sarimax import SARIMAX

def optimize_ARMA(endog: Union[pd.Series, list], order_list: list) -> pd.DataFrame:
  results = []
  for order in tqdm_notebook(order_list):
    try:
      model = SARIMAX(endog, order=(order[0], 0, order[1]),
              simple_differencing=False).fit(disp=False)
    except:
      continue
    aic = model.aic
    results.append([order, aic])
  result_df = pd.DataFrame(results)
  result_df.columns = ['(p,q)', 'AIC']

  #Sort in ascending order, lower AIC is better
  result_df = result_df.sort_values(by='AIC',
          ascending=True).reset_index(drop=True)
  return result_df
```

Example 3: ARMA(1, 1) — AIC

	(p,q)	AIC
0	(1, 1)	2801.407785
1	(2, 1)	2802.906070
2	(1, 2)	2802.967762
3	(0, 3)	2803.666793
4	(1, 3)	2804.524027
5	(3, 1)	2804.588567
6	(2, 2)	2804.822282
7	(3, 3)	2805.947168
8	(2, 3)	2806.175380
9	(3, 2)	2806.894930
10	(0, 2)	2812.840730
11	(0, 1)	2891.869245
12	(3, 0)	2981.643911
13	(2, 0)	3042.627787
14	(1, 0)	3207.291261
15	(0, 0)	3780.418416

- Fit all unique ARMA models:

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```

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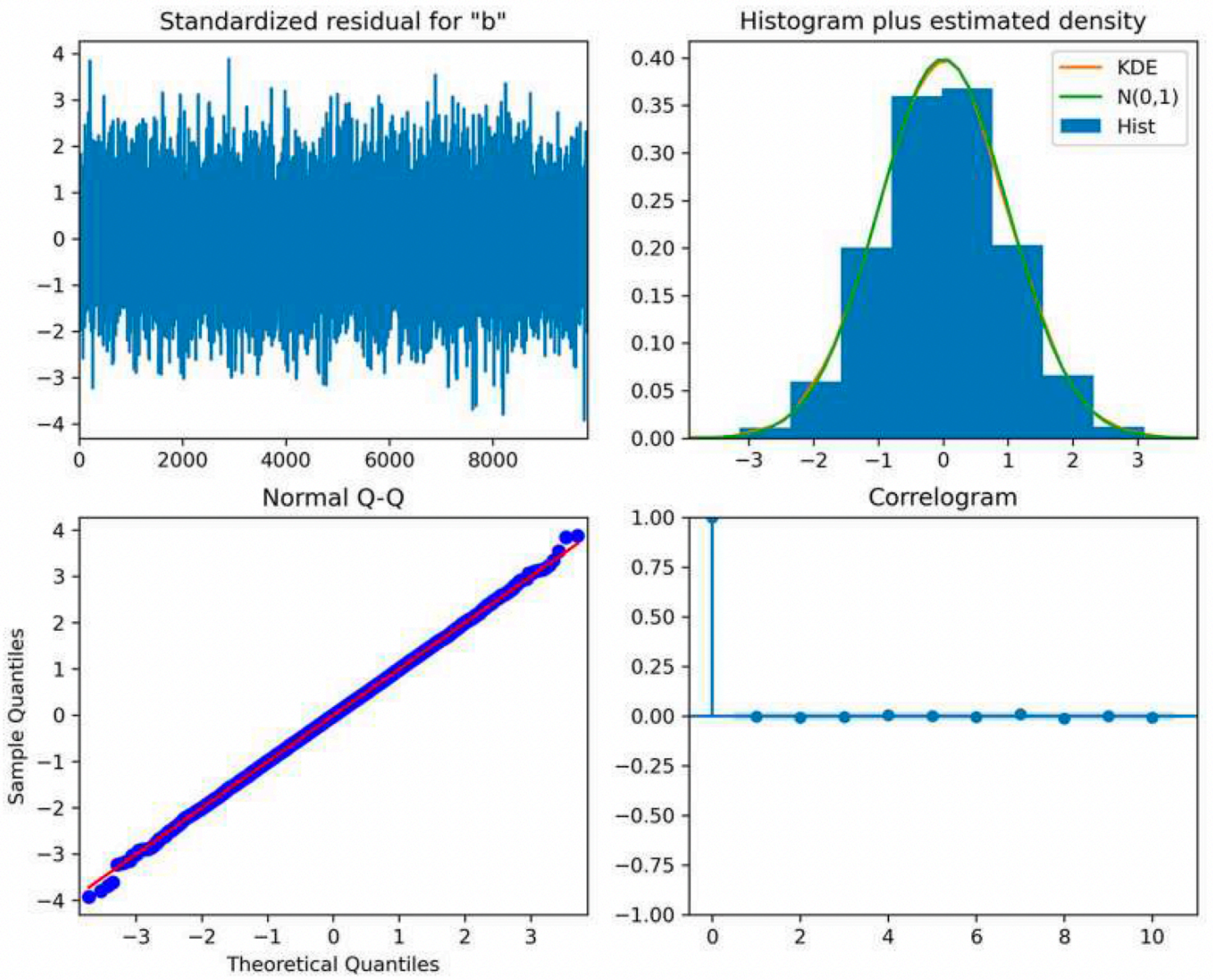
04

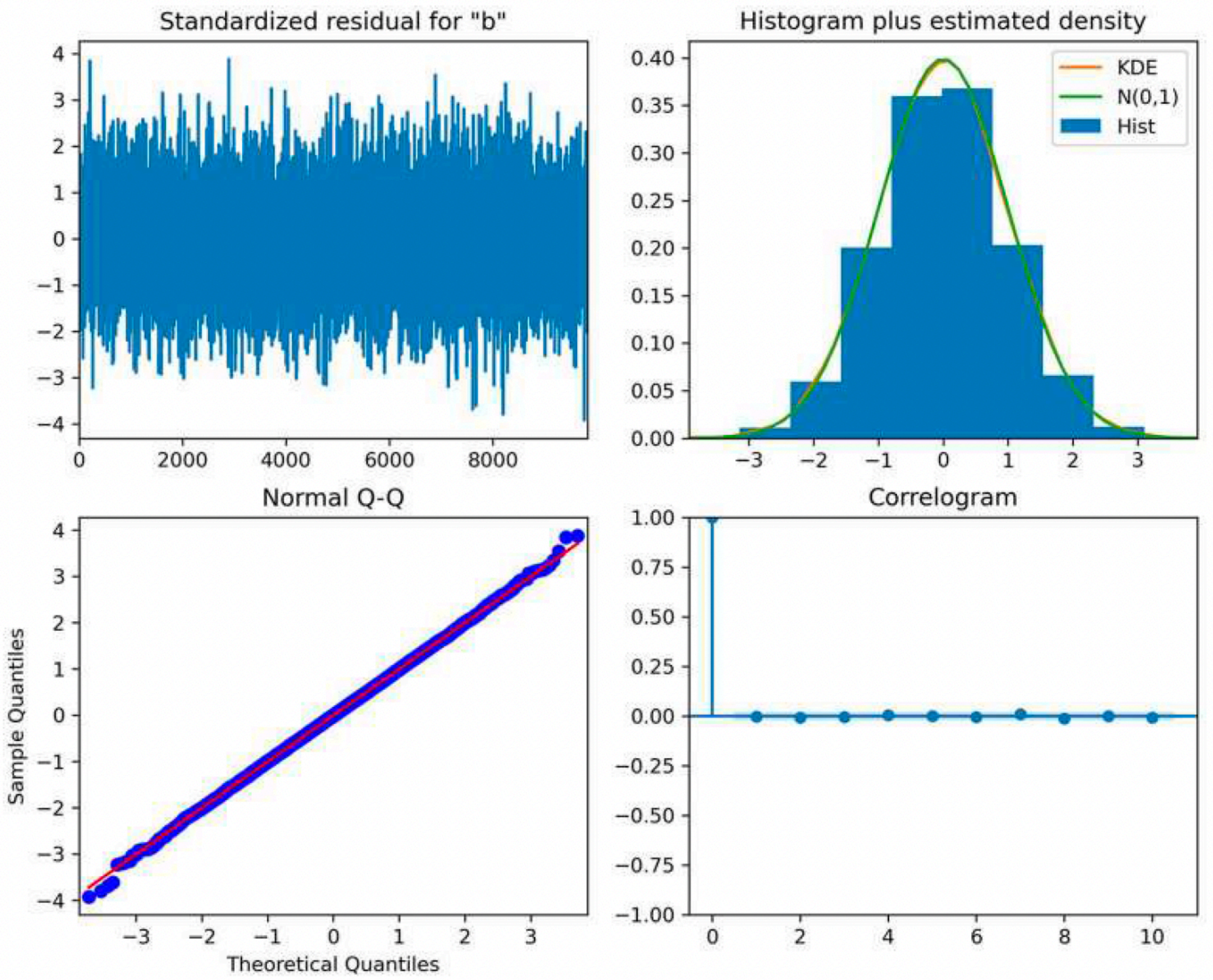
Residual Analysis

Residual Analysis



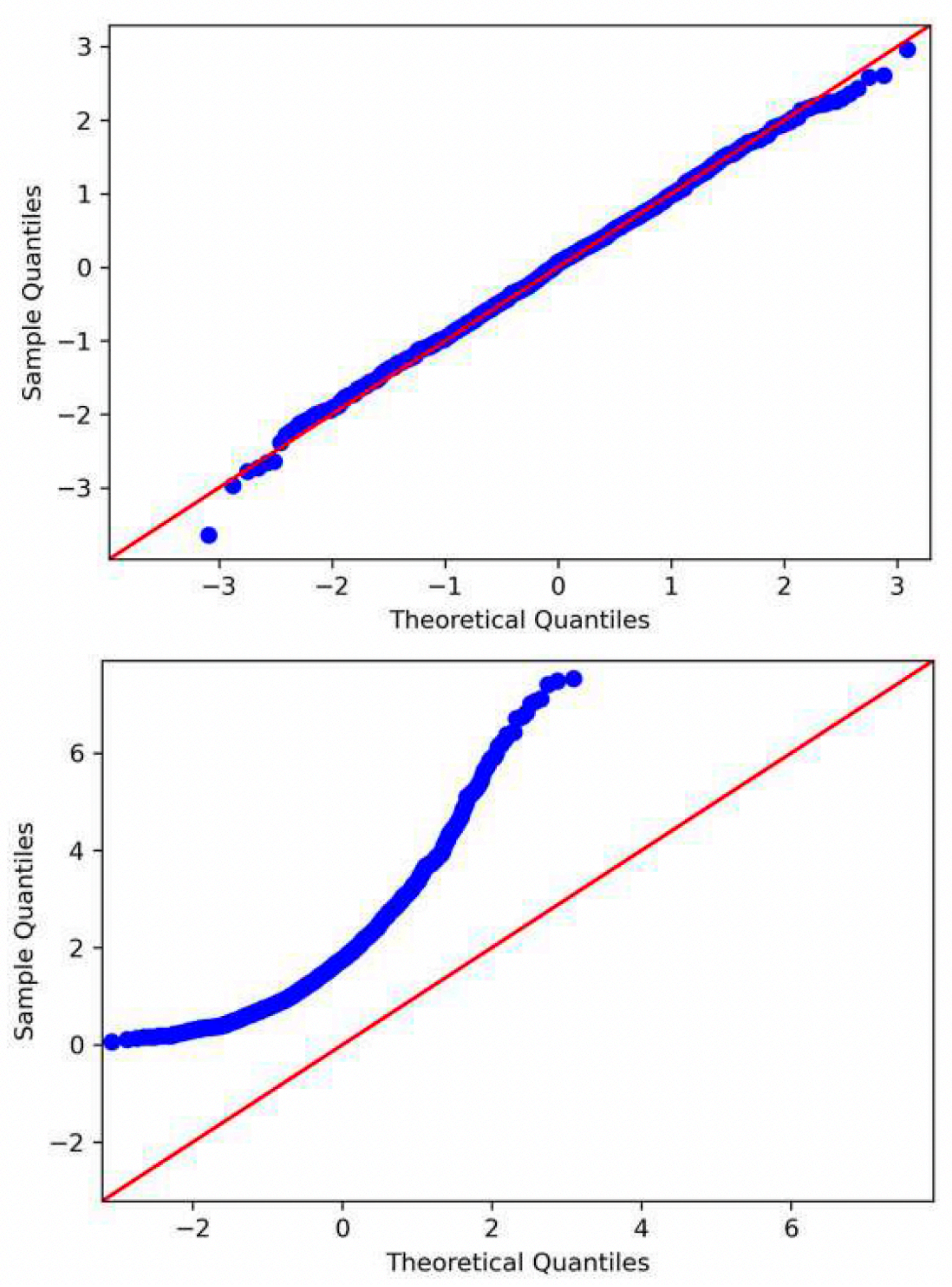
- Using the AIC as a model selection criterion, we found that an ARMA(1,1) model is the best model relative to all others that were fit.
- Now we must measure its absolute quality by performing an **analysis on the model's residuals**.
- The **residuals** are the differences between the values coming from our model and the observed values from our simulated process.
- The residuals must be **uncorrelated** and have a **normal distribution** in order for us to conclude that we have a good model for making forecasts.

Residual Analysis: Two Aspects



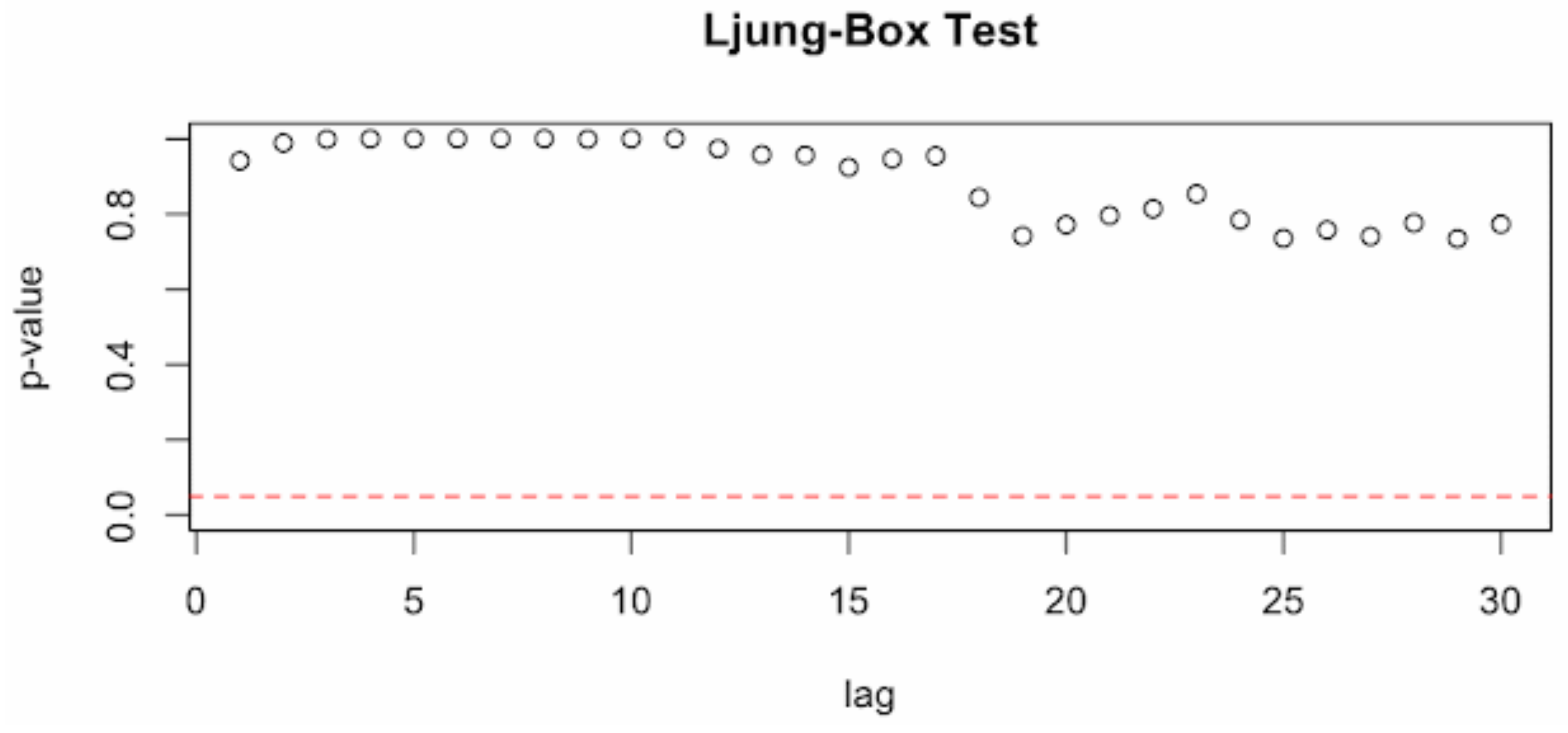
- There are two aspects to residual analysis: a qualitative analysis and a quantitative analysis.
- The qualitative analysis focuses on studying the **Q-Q plot**, while the **quantitative analysis** determines whether our residuals are uncorrelated.

Residual Analysis: Q-Q plot



- The **quantile-quantile plot (Q-Q plot)** is a graphical tool for verifying our hypothesis that the model's residuals are normally distributed.
- The Q-Q plot is constructed by plotting the quantiles of our residuals on the y-axis against the quantiles of a theoretical distribution, in this case the normal distribution, on the x-axis.
- If both distributions are similar, meaning that the distribution of the residuals is close to a normal distribution, the Q-Q plot will display a straight line that approximately lies on $y = x$. This in turn means that our model is a good fit for our data.

Residual Analysis: Ljung-Box Test



- The **Ljung-Box test** is a statistical test that determines whether the autocorrelation of a group of data is significantly different from 0.
- We apply the Ljung-Box test on the model's residuals to test whether they are similar to white noise.
- The *null hypothesis* states that the there is no autocorrelation.
- If the **p-value is larger than 0.05**, then the residuals are independently distributed.

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Thank you!